RNALigGUI User Manual

1. Introduction

RNALigGUI is a web-based platform for RNA–ligand binding affinity prediction and molecular dynamics (MD) trajectory analysis using the RNALig machine learning framework. The platform enables users to predict binding free energies directly from three-dimensional RNA–ligand complex structures and to analyze conformational ensembles generated during molecular dynamics simulations.

Key Features

- RNA–ligand binding affinity prediction
- Support for PDB and mmCIF structures
- Batch processing of multiple structures
- ZIP archive upload for MD trajectory analysis
- Frame-wise binding affinity prediction
- Average binding affinity estimation (RNALig_Frames)
- Interactive 3D molecular visualization
- Downloadable results

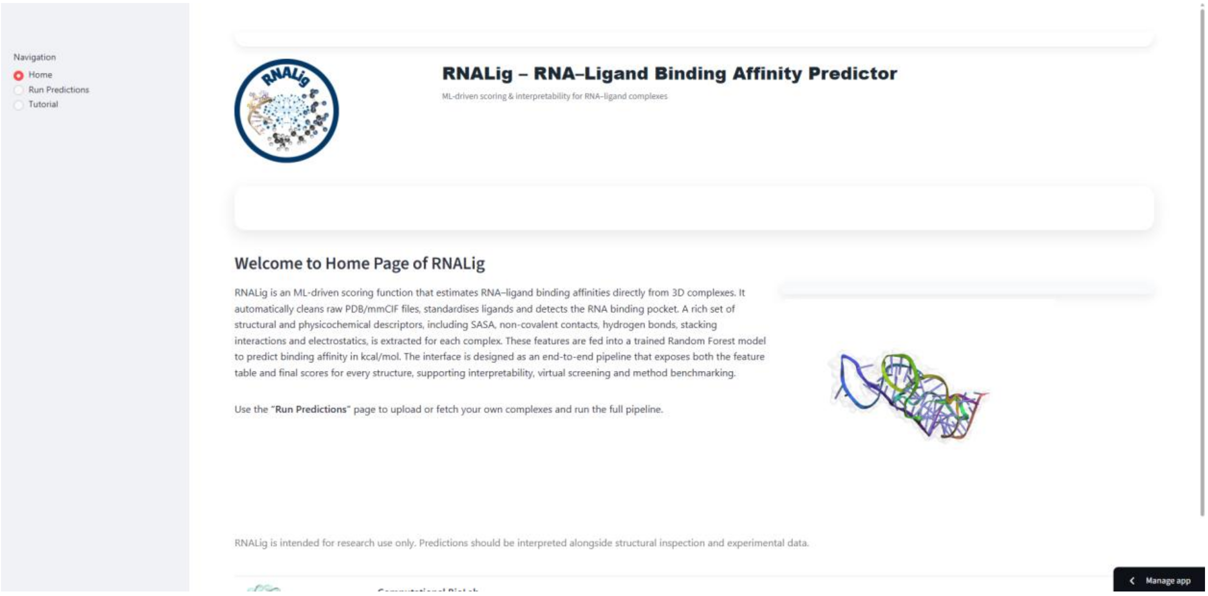
2. Accessing RNALigGUI

RNALigGUI is accessible through the web browser: https://rnalig-gui.streamlit.app/

No software installation is required.

3. Home Page

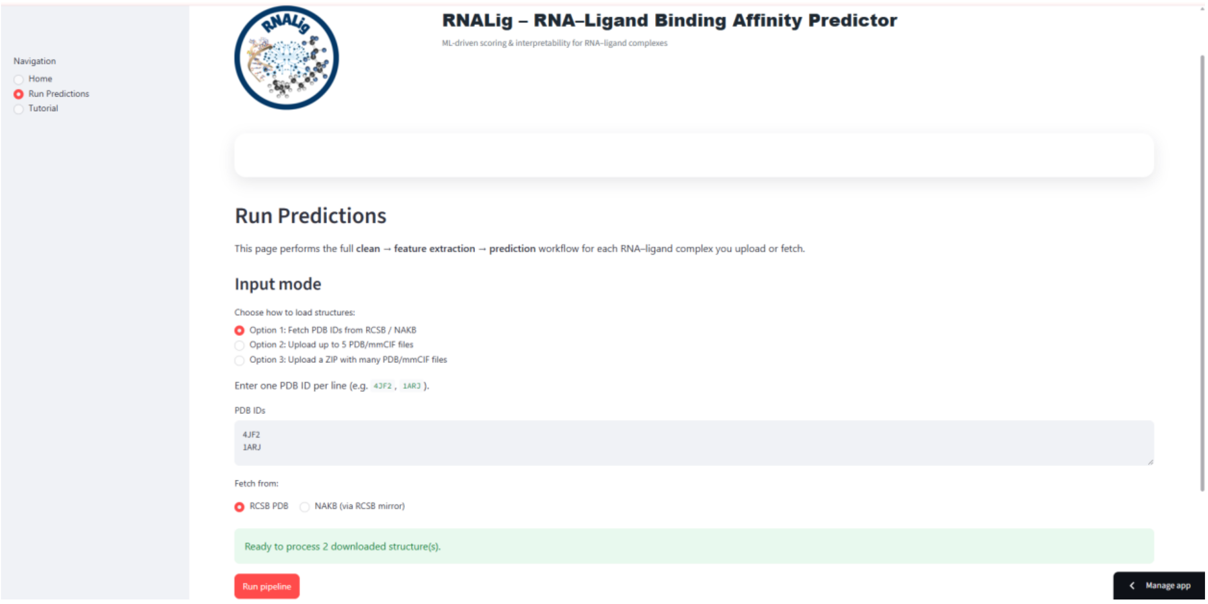
The home page provides an overview of the RNALig workflow and available prediction options.



Users can navigate to:

- Single Structure Prediction
- Batch Prediction
- MD Trajectory Analysis
- Documentation

4. Input Options



RNALigGUI supports three input modes.

Option 1: PDB Download

Users may enter one or more RNA–ligand complex PDB IDs.

Example:

4JF2

1ARJ

RNALigGUI automatically downloads the structures from the Protein Data Bank.

Option 2: Upload PDB/mmCIF Files

Users can upload up to five structures simultaneously.

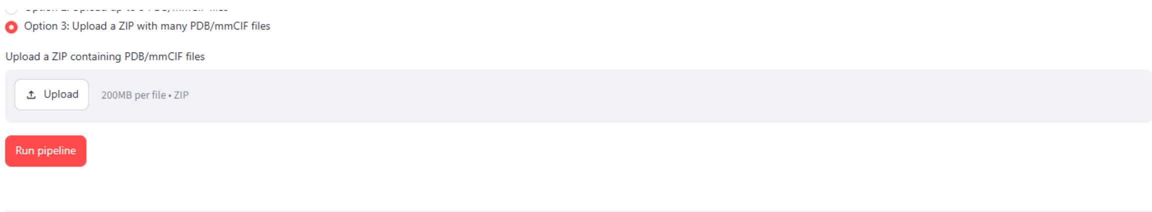
Supported formats:

- .pdb
- .cif

Option 3: Upload ZIP Archive

Users can upload a ZIP archive containing multiple RNA–ligand complex structures.

This option is particularly useful for molecular dynamics trajectory analysis.



5. Running Predictions

After selecting the desired input mode:

1. Upload structures or provide PDB IDs.
2. Click the "Run Prediction" button.
3. Wait for preprocessing and feature extraction to complete.
4. Review the prediction results.

6. Prediction Workflow

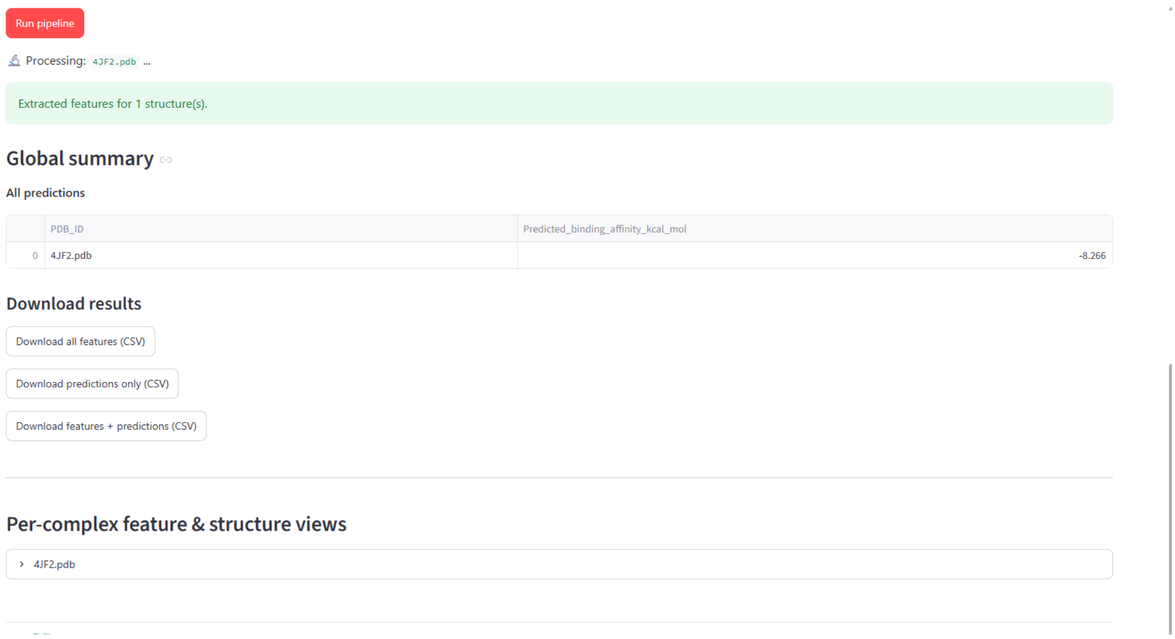
RNALigGUI performs the following operations automatically:

1. Structure preprocessing
2. RNA and ligand identification

3. Feature extraction
4. Descriptor generation
5. Random Forest prediction
6. Result generation

Workflow:

Raw RNA–Ligand Structure → Feature Extraction → RNALig Prediction → Binding Affinity Estimation



7. Understanding the Output

RNALigGUI provides:

Predicted Binding Affinity

Estimated binding free energy (ΔG) in kcal/mol.

Example:

Predicted Binding Affinity: -8.23 kcal/mol

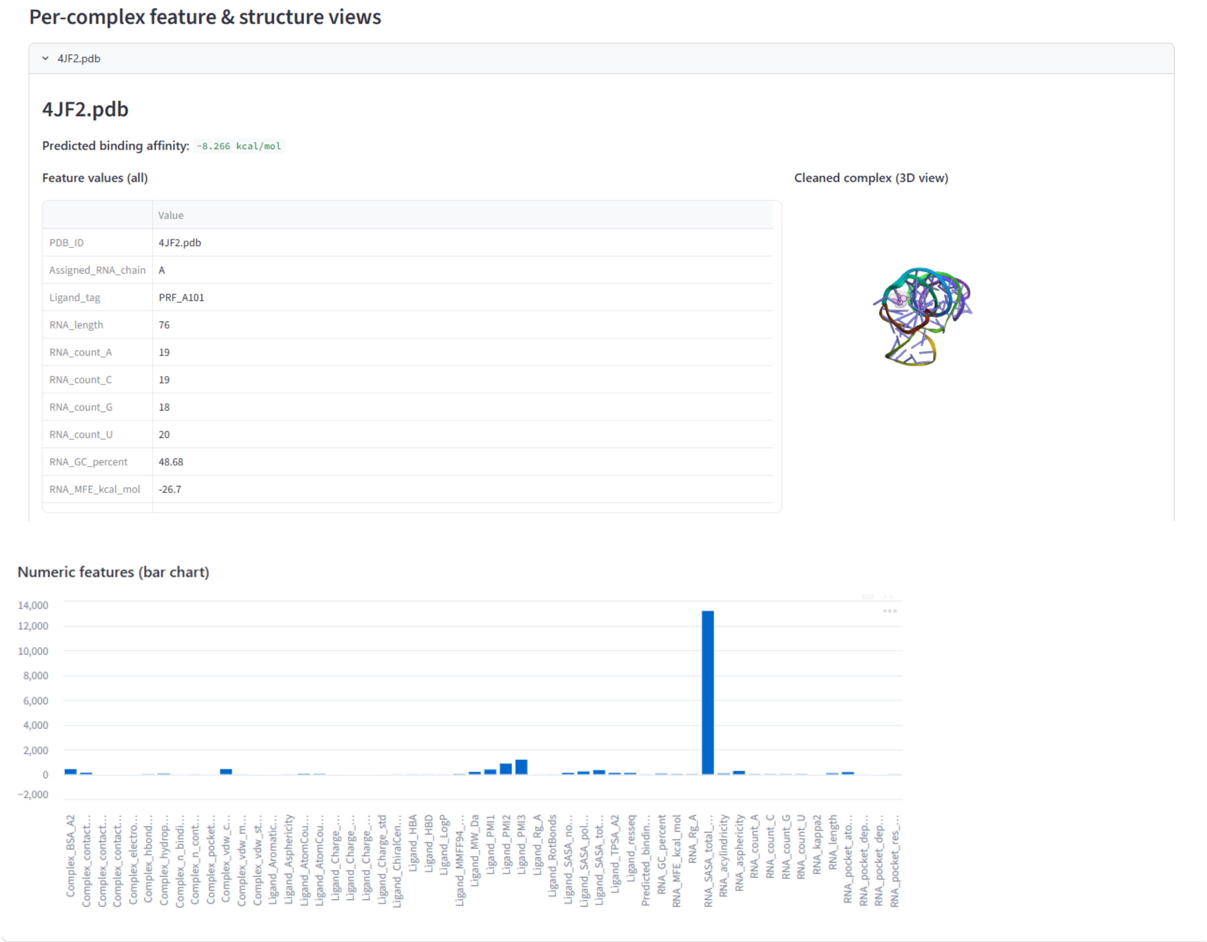
Structural Features

- RNA atom counts
- Ligand atom counts
- Hydrogen bonds
- Van der Waals contacts

- Stacking interactions
- Electrostatic interactions

Interactive Visualization

Users can inspect processed RNA–ligand complexes in a 3D viewer.



8. Molecular Dynamics Trajectory Analysis

RNALigGUI supports analysis of molecular dynamics trajectories through ZIP archive uploads.

Procedure

1. Extract frames from MD simulations.
2. Store all frames as individual PDB files.
3. Compress the files into a ZIP archive.
4. Upload the ZIP archive.
5. Run the prediction pipeline.

RNALigGUI predicts binding affinity for every frame.

The resulting frame-wise predictions can be used to:

- Monitor affinity fluctuations
- Assess trajectory stability
- Calculate average binding affinity (RNALig_Frames)
- Investigate temporal evolution of RNA–ligand interactions

9. Downloading Results

Users may download:

- Prediction tables
- Feature tables
- Binding affinity summaries
- Processed structures

Results are provided in CSV format.

10. Example Case Study

Target:

4JF2

Predicted Binding Affinity:

-8.26 kcal/mol

The platform automatically extracts descriptors and generates the final affinity estimate.

Global summary

All predictions

	PDB_ID	Predicted_binding_affinity_kcal_mol
0	4JF2.pdb	-8.266

11. Troubleshooting

Upload Failure

Ensure that:

- File format is supported.
- File size limits are not exceeded.

No Ligand Detected

Verify that the uploaded structure contains both RNA and ligand coordinates.

Invalid ZIP Archive

Ensure that all files inside the ZIP archive are valid PDB or mmCIF structures.

12. Citation

If you use RNALigGUI in your research, please cite:

Sharma P., Gupta T., Latha N., Pant P.

RNALig Web Server: An ML-Driven Platform for RNA–Ligand Binding Affinity Prediction and MD Trajectory Analysis.

13. Contact

Priyanka Sharma

NextGen Computational Biology Lab

Bennett University

India